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Ethan Arnold NE 533 Exam 4

$T = 625 \text{ K}, t = 500 \text{ days}, t^0 = 600 \mu\text{m}$

a) $t^* = 6.62 \times 10^{-7} \exp\left(\frac{11999}{625}\right) = 133 \text{ days}$ $t > t^*$

$\delta^* = 5.1 \exp\left(-\frac{550}{625}\right) \rightarrow \delta^* = 2.12 \mu\text{m}$

$K_L = 7.48 \times 10^{-6} \exp(-12500/625) \rightarrow K_L = 0.015 \frac{\mu\text{m}}{\text{d}}$

$\delta = 2.12 + (0.015)(500 - 133 \text{ days}) = 7.77 \mu\text{m}$

b) $PBR = 1.56, F = 0.14, \rho_{ZrO2} = 5.68 \text{ g/cc}, \rho_{Zr} = 6.5 \text{ g/cc}$

$$C_H = \frac{2(0.14) \times (7.77 \mu\text{m}) (5.68 \text{ g/cc}) \left(\frac{22}{32+16}\right) \times \left(\frac{1}{16}\right) \times 10^6}{(600 \mu\text{m} - \frac{2.12 \mu\text{m}}{1.56}) \times (6.5 \text{ g/cc})}$$

$C_H = 51.95 \text{ ppm}$ no initial H ppm given

2. Diffusion of Oxygen through the oxide is the rate limiting step

3. Hydrides form near the rim of the cladding. This is due to the lower temperature at the rim (Soret effect). Plate hydrides also preferentially form in higher tensile stress areas. Hydrides cause cladding embrittlement. Dry fuel can experience radial hydride reorienting leading to clad failure.

4. An RIA is a reactivity initiated accident where reactivity changes rapidly. (Control rod ejection (PWR) or control blade drop (BWR) occurs. During an RIA, a control rod is rapidly ejected and reactivity promptly increases (0.1s). In early stages, the gap closes, hydride pickup increases, and PWR failure may occur. During middle stages, DNB may occur where rapid coolant heating causes vapor layers on the cladding. Oxidation increases, the cladding becomes more ductile, and increased FGR may cause cladding bursts.

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4. In late stages, grains separate leading to increase FGR and potentially fuel fragmentation and relocation. If the clad bursts, the fuel may release to the coolant leading to blockages or damage from mechanical energy.

PWR: limiting at 0 power, normal P, T, rejection occurs from Pys when ROM fails

BWR: limiting at 0 power, at P, T-room, drop occurs from drive mech failure

5. A LOCA is a loss of coolant accident. It occurs when coolant stops being delivered to the core. This causes severely increased temperatures leading to increased oxidation and hydride pickup. Oxidation may hit breakaway states where H_2 is produced to unsafe quantities. The cladding typically balloons as system pressure no longer balances the internal clad pressure. Burst may occur and fuel fragmentation, relocation, and dispersal may occur, when the clad is removed from the ECCS, clad failure may occur from radial hydriding. Additionally, temperatures can get high enough to cause Zr to change to β or $\alpha+\beta$ phase which can demonstrate superplasticity leading to ballooning and rupture. LOCA mainly differs in time from RIA in that it is much less rapid. RIA also typically maintains coolant flow unless some function of fuel failure cuts this off locally.

6. Burnup affects fuel failure due to pellet specific enthalpy. Fresh fuel has much higher pellet specific enthalpy. However, burned fuel can see a greater increase in pellet specific enthalpy leading to failure of the cladding. Burnup also affects corrosion (initial oxidation is higher for burned fuel) and may affect fuel fragmentation (more cracks in burned fuel).

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7. Improved kinetics with steam/water, increase fuel mechanical properties, increase cladding mechanical properties, increased fission product retention
Ex: clad coating, pellet doping, larger pellet grains

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8. Addition of coatings to cladding is a near term concept for ATR (Co, Al). These coatings drastically reduce oxidation (and hydrogen pickup) and create more stable passivated layers on the cladding.
- steam specifically

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9. Zr can oxidize rapidly under high temperature steam. It also experiences more hydride pickup and becomes more brittle while seeing thinned walls. The excess H_2 is also a flammable risk (Fukushima). Breakaway oxidation can occur and feedback loops (higher T $\rightarrow$ more oxide thickness $\rightarrow$ higher T) may occur. There may also be risk of delayed hydride cracking later.
- phase changes

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10. DNB - Departure from Nucleate Boiling (shall not occur under normal operation)
Corrosion - Oxide layer less than 17% of clad thickness

RIP - Rod internal pressure, no clad lift-off to maintain heat transfer

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11. CRUD (or Chalk River Unidentified Deposit) can lower thermal conductance between the clad and the coolant, cause higher clad temperature, lead to increased oxidation, and cause CIPS (Circuit Induced Power Shift).

CIPS occurs as boron is deposited in the crud (in upper grid spaces where subcooled Nucleate Boiling occurs). This pushes the axial power peak down the assembly and can degrade shutdown margin.
- radiation source

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12. The addition of $LiOH$ to water maintains the pH of the coolant above 7 and decreases the deposition of CRUD. $LiOH$ is added to counter boric acid (used for reactivity control). Zinc is also added to reduce activated crud deposition on core surfaces and allow

cleanup, thereby reducing dose to plant workers.

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13. Max fuel forms 4 distinct regions: a central void, columnar grain region, equiaxed grain region, and as-sintered region. The high power density of max fuel leads to formation of gaseous pores in the microstructure. These pores travel up the temperature gradient by evaporating and condensing fuel (due to the high T gradient across the low thermal conductivity pores). The pores coalesce in the center of the fuel pellet leading to the formation of the central void. Additionally, SiO_2 (usually Si_2MoO_4) forms between the pellet and clad surface due to the roughness of the surface not fully closing the gaps. SiO_2 may reach 150 μm in thickness. It has a much lower k than the fuel but higher than fission gas. It is also fairly soft reducing the likelihood of mechanical PCMI failure. Finally, it allows for a reduction of fission product swelling in the fuel by allowing release of fission products to a more chemically favorable state. Max fuels also have significantly more fission products released into the microstructure. There are visibly more ~ 5 metal precipitates.